## Investigation: BAHEMA SA (CVM code 701), fiscal year 2011, as it was on 2016-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Emp. Adm. Part. - Sem Setor Principal (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

## Document used

2011 DFP, version 1, filed on 2012-03-16; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2012-03-16).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=15069

## Possible manipulation (engine v1, study 1)

## Financial risk (context only)

Altman Z'' (financial risk, context only)
  Value: 12.060 (index)
  Components: X1 = 0.847; X2 = 0.384; X3 = −0.060; X4 = 5.387 (indices)
  Percentile in sector and year: 80 (n = 76)
  The company's own history: fewer than 5 previous fiscal years with a value
  - No cut-off zone: the available source does not define zones for Z''
  - Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
  Value: F = 3/9 (7 computable)
  Signals:
    - Return on assets (ROA): 0
    - Operating cash flow on assets (CFO): 1
    - Change in ROA: 0
    - Cash above earnings (ACCRUAL): 1
    - Change in leverage: 0
    - Change in current ratio: 1
    - No equity issuance (EQ_OFFER): —
    - Change in gross margin: —
    - Change in asset turnover: 0
  - No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement
  - No long-term debt in either year: the change in leverage is zero and the signal is 0 (project choice; the article only defines fell = 1, rose = 0, p. 8)

## Changes

Highlighted in the following year's comparative: none.
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets

of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 11 accounts above the R$1,000 tolerance (group a, totals and key lines: 2; group b, reclassifications: 9), in the annex at the end.

## What was known on the date

- DFP 2010, v1: 2011-03-04
- FCA 2011, v1: 2011-03-04
- FCA 2011, v2: 2011-05-10
- DFP 2011, v1: 2012-03-16
- DFP 2012, v1: 2013-04-01

DFP and FCA filed with the CVM by the reference date; ITR comes in a later phase.
Material facts and market notices filed by the date (IPE open data, since 2010): 8 material fact documents and 13 market notice documents.
Material facts filed in the 24 months before the date (from 2014-05-01 to 2016-04-30):

- 2016-04-29: Plano de Outorga de Opção de Compra de Ações
- 2015-05-01: Renúncia de membros do Conselho de Administração
- 2014-07-31: Alteração do estatuto social||Reorganização Societária - Incorporação, Cisão ou Fusão da Companhia
- 2014-07-15: Incorporação de controlada
- 2014-07-01: Início dos estudos para incorporação de controlada

Older material facts, the market notices and the link to each document are in the JSON and on screen.
Material facts and market notices filed with the CVM by the date, from the IPE open data (since 2010): filing date, category, subject (as filed, in Portuguese) and link. The content of the documents is not downloaded, read or summarized, and the list generates no alert, score or ranking. The counts are of documents: versions, resubmissions and English versions count separately; in the list, versions of the same document and documents with the same date, category and subject appear on a single line, with a link to each one.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2013 FCA: Audilink & Cia.Ltda.
Auditor in the 2014 FCA: Audilink & Cia.Ltda.
Auditor in the 2015 FCA: BKS Auditores.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2015: Audilink & Cia.Ltda. -> BKS Auditores. The outgoing firm audited 5 consecutive fiscal years (2010 to 2014), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## Document excerpts (copied; no alert)

Excerpts copied from the documents, with the page, and the Reference Form table. It is not a signal family: it generates no alert, score or ranking. Absence of mention does not prove absence.

How the reader was tested: 22 held-out cases, a single measurement with the frozen reader (dd688da): precision 0.973, coverage 0.915, literal check on 146 of 146 excerpts, no natural-person name (docs/leitor_medida_f1_2.json).

Explanatory notes · Related parties
Note: NOTA 9. PARTES RELACIONADAS (pages 8 to 8)
- [subsidiaries] “Débitos com controladas 12.396 12.396” (p. 8, original excerpt in Portuguese)
- [parent company; subsidiaries] “A controladora é credora da controlada de um mútuo. O prazo de pagamento do mutuo é por” (p. 8, original excerpt in Portuguese)

Fields of item 18 of CPC 05 (R1)
- Excerpt omitted; see page 8 of the document
- [balances] “Débitos com controladas 12.396 12.396” (p. 8, original excerpt in Portuguese)
- [amount of the transactions] “RESULTADO RECEITA FINANCEIRA 0” (p. 8, original excerpt in Portuguese)
- [terms and conditions] “Juros 1.309 1.309” (p. 8, original excerpt in Portuguese)
- [amount of the transactions; terms and conditions] “A controladora é credora da controlada de um mútuo. O prazo de pagamento do mutuo é por” (p. 8, original excerpt in Portuguese)
- [terms and conditions] “tempo indeterminado e incide encargos de 100% do CDI mensal e juros de 0,30% am,” (p. 8, original excerpt in Portuguese)
- [balances] “capitalizados mensalmente sobre o saldo devedor. Os valores vêm sendo amortizados conforme as necessidades de caixa das empresas.” (p. 8, original excerpt in Portuguese)

Explanatory notes · Reverse factoring and forfaiting
The reader found no mention of reverse factoring (risco sacado), forfaiting or supplier agreements in the notes of this DFP. Absence of mention does not prove absence of the operation.

Auditor's report
Type of opinion: unqualified
- Excerpt omitted; see the auditor's report in the document

Reference Form
Form of 2012, version 5, filed on 2013-04-03.
It is the only version available in the open data; earlier versions are not there.
- Related party (name omitted) · Controlada · empréstimo de mútuo · 12396170.56 · 12396170,56

Limits of this section:
- In real documents, about 22% of the excerpts come out omitted by the name detector, including names of companies outside the registry.
- In tables, a header line (years, column titles) may still come out as an excerpt: in the reader's measurement, 1 of 19 header texts of the held-out cases came out as excerpts.

## How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a F1); leitor-f1-2 (commit dd688da). The generation date is in the demo manifest (demo/manifesto.json).

## Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: BIOMA EDUCAÇÃO S.A.

## Annex: full list of differences in the following year's comparative (R$ million)

(a) BPP 2.01 Passivo Circulante: 1.6 in the version used, 1.7 in the comparative (difference of 0.1)
(a) BPP 2.02 Passivo Não Circulante: 9.0 in the version used, 8.9 in the comparative (difference of −0.1)
(b) DFC (indirect method) 6.01.01 Caixa Gerado nas Operações: 8.2 in the version used, 5.4 in the comparative (difference of −2.8)
(b) DFC (indirect method) 6.01.03 Outros: −2.8 in the version used, 0.0 in the comparative (difference of 2.8)
(b) DFC (indirect method) 6.01.03.01 LUCRO LÍQUIDO / (PREJUÍZO) DO EXERCÍCIO: −2.8 in the version used, 0.0 in the comparative (difference of 2.8)
(b) BPP 2.01.03 Obrigações Fiscais: 0.9 in the version used, 1.0 in the comparative (difference of 0.1)
(b) BPP 2.01.03.01 Obrigações Fiscais Federais: 0.9 in the version used, 1.0 in the comparative (difference of 0.1)
(b) BPP 2.01.03.01.02 Obrigações Fiscais Federais: 0.9 in the version used, 1.0 in the comparative (difference of 0.1)
(b) BPP 2.02.02 Outras Obrigações: 6.7 in the version used, 6.6 in the comparative (difference of −0.1)
(b) BPP 2.02.02.02 Outros: 6.7 in the version used, 6.6 in the comparative (difference of −0.1)
(b) BPP 2.02.02.02.04 Parcelamento Federal - REFIS ( Lei 11.941/09 ): 5.7 in the version used, 5.6 in the comparative (difference of −0.1)